

Rahul Anand Sharma

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| Mountain View, CA

SUMMARY

Staff / Lead Machine Learning Systems Engineer and Researcher specializing in hardware-software co-design, LLM inference acceleration, and distributed accelerator infrastructure (TPUs/GPUs). Ph.D. from CMU with 15+ publications at top systems venues (SIGCOMM, NSDI, USENIX Security) and 2 US patents. Track record of driving high-impact technical initiatives across Google Cloud (TPU v7 Ironwood NPI, Gemini disaggregated serving for Sovereign Cloud), AWS (Amazon Q / Generative AI at hyperscale), and Microsoft Research.

TECHNICAL SKILLS

Languages: Python, C++, CUDA, SQL, Bash

ML Frameworks & Compilers: JAX, PyTorch, XLA, Triton, TensorRT / TensorRT-LLM, TensorFlow

LLM Systems & Serving: Disaggregated Serving, vLLM, Model Parallelism (Tensor/Pipeline/Expert), Distributed Training (FSDP, Megatron-LM), KV-Cache Optimization, Speculative Decoding, Quantization (FP8/INT4)

Systems, Cloud & Hardware: Google Cloud TPUs (v4/v5/v7), NVIDIA GPU Clusters, Google Distributed Cloud (GDC), AWS (Glue, EMR, EC2), Apache Spark, Kubernetes, Docker, Ray

Core Domains: LLM Inference Acceleration, Multi-Cube Accelerator Scaling, Compiler-Hardware Co-design, Network & Systems Security

PROFESSIONAL EXPERIENCE

Google Cloud

Aug. 2025 – Present

Senior Machine Learning Engineer (MLE)

Mountain View, CA

- Led the TPU v7 (Ironwood) NPI microbenchmarking suite; bridged internal cluster infrastructure (Borg) with Cloud XLA runtimes, ported benchmarks from HLO to JAX, and executed multi-cube scaling analyses to model accelerator interconnect limits.
- Delivered collective communication microbenchmarks and public recipes for the TPU v7 public preview, establishing interconnect scaling baselines that advanced multi-billion-dollar hardware evaluations with tier-1 frontier AI customers.
- Spearheaded the intra-node (4:4 split) disaggregated serving architecture for Gemini models on bare-metal nodes in Google Distributed Cloud (GDC), overcoming strict memory isolation and control-plane constraints for Sovereign Cloud.
- Architected automated continuous regression pipelines ensuring Borg-Cloud runtime parity; optimized open-source LLMs (Llama 3, DeepSeek) to land Day-1 3P hardware enablement (Sim-Ship), earning the Google Cloud Tech Impact Award (2026).

AWS, Amazon

Jan. 2023 – July 2025

Applied Scientist

Palo Alto, CA

- Architected and launched a Generative AI-powered auto-debugging system (Amazon Q) for Apache Spark across AWS Glue, Amazon EMR, and MaxDome, utilizing LLMs for automated root-cause analysis to slash Mean Time to Resolution (MTTR) from days to minutes.
- Engineered a high-throughput active testing service for AWS network security, injecting synthetic live traffic to validate Network ACLs at scale, guaranteeing strict compliance and eliminating false-positive blindspots inherent to static simulation models.
- Designed and integrated automated diagnostic tooling directly into AWS Glue and EMR management consoles, delivering real-time failure insights and actionable remediations to thousands of enterprise customers.

Microsoft Research, Redmond

May 2020 – July 2021

Research Intern

Redmond, WA

- Architected low-latency distributed protocols for edge-cloud deep learning inference offloading within the Networking Research Group, minimizing communication overhead and end-to-end latency under bandwidth-constrained networks.

Microsoft Research, India

July 2016 – July 2018

Research Fellow

Bangalore, India

- Co-designed and built FarmBeats, an end-to-end IoT and cloud analytics platform for data-driven agriculture; awarded Microsoft Outstanding Technical Achievement Award and productized into Microsoft Azure (US Patent US20200150640A).
- Pioneered DeepEdge, a novel edge-AI scheduling architecture optimizing deep learning workloads under severe compute and bandwidth constraints, enabling graceful performance degradation (US Patent US10942767B2; published at NSDI).
- Researched neural network learnability with theoretical researchers, analyzing empirical bounds on how data complexity, model depth, and minibatch dynamics govern generalization.
- Mentored 4+ research interns and junior engineers, driving collaborative systems research initiatives that resulted in multiple conference publications.

EDUCATION

Carnegie Mellon University

Ph.D. in Electrical & Computer Engineering

- *Thesis:* Practical Network Layer Machine Learning for IoT Security
- *Advisors:* Prof. Vyas Sekar & Prof. Anthony Rowe | *Honors:* CIT Dean's Fellow

Pittsburgh, PA

Aug. 2018 – Jan. 2023

International Institute of Information Technology (IIIT)

B.Tech and M.S. by Research in Computer Science

- *Honors:* Dean's List for Academic and Research Excellence

Hyderabad, India

July 2011 – July 2016

PATENTS & HONORS

Patents: [US10942767B2](#) (Deep neural network workload scheduling); [US20200150640A1](#) (Sensor fall-curve identification)

Awards & Honors: Google Cloud Tech Impact Award (2026); Microsoft Outstanding Technical Achievement Award (FarmBeats, 2018); CIT Dean's Fellow, CMU (2018); Dean's List, IIIT Hyderabad (2016)

Media & Recognition: Research featured on the front page of Hacker News, The Washington Post, and NDTV.

SELECTED PUBLICATIONS

15+ peer-reviewed papers at top-tier systems, security, and networking venues with 450+ citations.

USENIX Security 2022 | *Lumos: Identifying and Localizing Diverse Hidden IoT Devices in an Unfamiliar Environment*

– Built a mobile system leveraging 802.11 PHY/MAC features and AR to localize hidden IoT devices with 95% accuracy; covered by Hacker News front page and CyLab.

ACM CoNEXT 2022 | *Lumen: A Framework for Developing and Evaluating ML-Based IoT Network Anomaly Detection*

– Designed an ML evaluation framework across 100+ devices, resolving domain-shift degradation in network traffic anomaly detection.

ACM SIGCOMM 2020 | *Contention-Aware Performance Prediction for Virtualized Network Functions*

– Developed an ML-driven contention-aware predictor for virtualized network functions (VNFs) to guarantee QoS under multi-tenant cloud workloads.

ACM IPSN 2020 | *All that GLITTERs: Low-Power Spoof-Resilient Light Anchors for Augmented Reality*

– Engineered an optical retro-reflective sensing pipeline enabling sub-centimeter AR localization at milliwatt power consumption.

USENIX NSDI 2019 | *DeepEdge: A Network Edge for Deep Learning Workloads*

– Designed a distributed edge-cloud scheduling framework optimizing DNN inference pipelines under tight compute constraints.

SELECTED INVITED TALKS

USENIX Security 2022 | “Lumos: Identifying and Localizing Diverse Hidden IoT Devices” (Boston, MA)

CyLab & CONIX Annual Review 2022 | “Lumen: ML-Based IoT Network Anomaly Detection” (CMU, Pittsburgh, PA)

ACM IPSN / CPS-IoT Week 2020 | “All that GLITTERs: Light Anchors for Augmented Reality”

Microsoft Techfest 2018 | “FarmBeats: AI, Edge & IoT for Agriculture” (Redmond, WA)

IEEE WACV 2018 | “Automated Top-View Registration of Broadcast Sports Video” (Lake Tahoe, NV)